SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1– Group Debate on AI Ethics, Bias, Fairness, Responsible AI

Course Code:	CSA65	**Course Title:**	Generative AI and Large Scale Models
CO Assessed:	CO5- BL-5 – Articulation	**Assessment:**	Assessment Tool 3 - Group Debate on AI Ethics, Bias, Fairness, Responsible AI
Weightage:	10% of CIA	**Total Marks:**	1 * 100= 100
CO4 Statement:	*Deploy Generative AI applications while evaluating ethical, security, governance, and responsible AI considerations in industrial environments.*		
Student Name:	V Rohan	**Reg. No.:**	192472227
Section / Batch:	CSE AI	**Date:**	31 / 08 / 26

Instructions to Students

- Team Size: 8 Members
- Each team will be assigned one debate topic.
- Debate Instructions
- Each team will have 8 members.
- Each team will be assigned one topic.
- Divide the team into two groups: For (4 members) and Against (4 members).
- Each member should present clear arguments with supporting evidence.
- Use real-world AI examples, case studies, research findings, or statistics wherever relevant.
- Each team should include opening arguments, rebuttals, and a concluding statement.
- All 8 members must actively participate in the debate.
- Maintain professional communication and respectful disagreement.
- Avoid unsupported claims or personal attacks.
- The team should conclude by stating the strongest arguments from both perspectives.
- Duration: 15–20 minutes per team.

Code of Conduct

I _V Rohan (192472227)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___Rohan V__________

SECTION A : Group Debate on AI Ethics, Bias, Fairness, Responsible AI

1. **"AI systems can never be completely free from bias."**

Team Structure

The 8-member team is divided into two groups of four. The "For" team argues that bias in AI systems is fundamentally unavoidable. The "Against" team argues that, while difficult, AI systems can be made effectively free from meaningful bias through rigorous technical and governance measures. Each side presents an opening argument, two evidence-based arguments, a rebuttal, and a joint concluding statement one role per speaker.

FOR THE MOTION "Bias Is Unavoidable" (4 Speakers)

Speaker 1 (For) - Opening Statement

Good [morning/afternoon] everyone. Our team stands firmly in support of the motion: AI systems can never be completely free from bias. Bias does not enter AI systems as an accident that can simply be patched out it enters through the data these systems learn from, the people who design them, and the objectives they are trained to optimize. Since AI is built by humans, trained on human-generated data, and deployed into unequal societies, some degree of bias is structurally embedded at every stage of the pipeline: data collection, labeling, model design, and real-world deployment. Over the next few minutes, my teammates and I will show that this is not a temporary engineering gap but a fundamental characteristic of how machine learning works.

Speaker 2 (For) - Evidence-Based Argument I: Bias in Training Data

Consider the COMPAS recidivism-risk tool used in several U.S. courts, which an investigative analysis by ProPublica found flagged Black defendants as future re-offenders at nearly twice the rate of white defendants with similar records. The algorithm was not explicitly told a defendant's race, yet it absorbed racial disparities already present in historical arrest and sentencing data. Similarly, Joy Buolamwini and Timnit Gebru's "Gender Shades" study found commercial facial-recognition systems from major technology companies had error rates below 1% for light-skinned men but above 30% for dark-skinned women, because training datasets were overwhelmingly composed of lighter-skinned faces. These are not edge cases they show that whenever historical human decisions shape training data, the model inherits the inequities embedded in that history.

Speaker 3 (For) - Evidence-Based Argument II: Bias in Design and Deployment

Bias also emerges from choices made by system designers, not just data. Amazon's internal AI recruiting tool, scrapped in 2018, penalized résumés containing the word "women's" (as in "women's chess club") because it had learned from a decade of male-dominated hiring patterns even after engineers tried to neutralize it. Every design decision which features to include, how to define "success," which population to test on reflects the values and blind spots of the people making it. And even mathematically, computer scientists have proven through the impossibility theorems in fairness literature (e.g., Kleinberg, Mullainathan & Raghavan, 2016) that multiple reasonable definitions of fairness such as equal false-positive rates and equal predictive accuracy across groups cannot all be satisfied simultaneously except in trivial cases. This means any AI system inevitably makes a trade-off that disadvantages some group.

Speaker 4 (For) - Rebuttal & Response to Opposition

Our opponents will likely argue that techniques such as re-weighting datasets, adversarial de-biasing, or fairness toolkits like IBM's AI Fairness 360 can eliminate bias. We agree these tools reduce bias but reduction is not elimination. Every de-biasing technique optimizes for one specific, chosen definition of fairness, and as shown by the impossibility results, choosing one definition necessarily sacrifices another. Furthermore, new bias is continuously reintroduced as models are retrained on fresh real-world data, deployed in new contexts, or used by populations the original developers never anticipated. "Completely free from bias" is a static, absolute claim; what the opposition can realistically offer is only "bias reduced to an acceptable level" which is a different and much weaker claim than the one the motion requires them to refute.

Speaker (For) - Concluding Statement (delivered jointly / by Speaker 1 or 4)

To summarize: bias enters AI through historical data (COMPAS, Gender Shades), through human design choices (Amazon's hiring tool), and through mathematically provable trade-offs between competing fairness definitions. Mitigation is possible and necessary but complete elimination is not, because bias is woven into the very process by which these systems learn from an unequal world. For these reasons, we strongly affirm the motion.

AGAINST THE MOTION - "Bias Can Be Effectively Eliminated" (4 Speakers)

Speaker 1 (Against) - Opening Statement

Good [morning/afternoon]. Our team respectfully opposes the motion. While we acknowledge that early and poorly-governed AI systems have shown serious bias, the claim that bias can "never" be eliminated is an overly pessimistic and, frankly, outdated view. The field of Responsible AI has matured rapidly: today we have mature fairness-testing frameworks, regulatory standards, and technical tools specifically designed to detect and neutralize bias before and after deployment. We will show that with rigorous auditing, diverse data governance, and continuous monitoring, AI systems can reach a standard of fairness that is functionally indistinguishable from "free from bias" for real-world purposes and that treating bias as an unsolvable inevitability discourages the very investment needed to fix it.

Speaker 2 (Against) - Evidence-Based Argument I: Technical De-biasing Works

Tools such as IBM's AI Fairness 360 toolkit and Google's What-If Tool allow engineers to measure and correct disparities such as demographic parity and equalized odds before a model ever reaches production. After the Gender Shades study exposed facial-recognition bias, IBM, Microsoft, and other developers retrained their systems on more diverse datasets and, in follow-up audits, reported dramatically reduced error-rate gaps across skin tones. This demonstrates that bias is not a fixed law of nature but a measurable, correctable engineering defect one that shrinks substantially once it is identified and targeted, just as security vulnerabilities are patched once discovered.

Speaker 3 (Against) - Evidence-Based Argument II: Governance and Regulation Close the Gap

Beyond technical fixes, structured governance closes the remaining gap. The EU AI Act mandates bias and conformity testing for "high-risk" AI systems before they can be deployed, and the U.S. NIST AI Risk Management Framework provides standardized auditing practices that organizations increasingly follow. Amazon's own decision to scrap its biased hiring tool in 2018, rather than deploy it, is itself proof that governance processes can catch and stop biased systems the process worked as intended. When technical de-biasing is paired with mandatory external audits, diverse review boards, and continuous post-deployment monitoring, residual bias can be pushed down to a negligible, practically "bias-free" level, especially compared to the well-documented, unregulated biases of human decision-makers these systems often replace.

Speaker 4 (Against) - Rebuttal & Response to Opposition

Our opponents cite the impossibility theorems in fairness literature to claim bias is mathematically guaranteed. But those theorems show that multiple fairness definitions cannot all be satisfied at once they do not prove that a well-chosen, context-appropriate fairness metric cannot be satisfied to a practically complete degree. A cardiac surgeon cannot guarantee a zero-risk operation in the absolute sense, yet we still call well-executed surgery “safe.” Likewise, when a system is audited, monitored, and corrected on an ongoing basis using the most contextually relevant fairness metric, the remaining bias becomes so small and so rapidly correctable that it no longer meaningfully differs from “completely free from bias” in practice. Treating imperfection as impossibility sets an unreachable standard and, worse, can be used to excuse organizations from continuing to invest in fixing the bias that remains.

Speaker (Against) - Concluding Statement (delivered jointly / by Speaker 1 or 4)

To summarize: fairness toolkits (AI Fairness 360, What-If Tool), documented retraining successes after Gender Shades, and binding governance frameworks (EU AI Act, NIST AI RMF) together show that bias is a solvable engineering and governance problem, not an unavoidable fate. Progress already made including Amazon halting its own biased tool shows the system of detection and correction functions. For these reasons, we oppose the motion.

Joint Closing - Strongest Arguments from Both Sides

(To be delivered by any nominated member from the team, summarizing both perspectives as required by the assessment instructions.)

- Strongest “For” argument: Documented cases (COMPAS, Gender Shades, Amazon's recruiting tool) and mathematical impossibility results show bias is structurally embedded in data, design, and competing fairness definitions not merely a bug.
- Strongest “Against” argument: Technical fairness tools and binding regulatory frameworks (EU AI Act, NIST AI RMF) have measurably reduced bias in real deployments, showing that bias is manageable to a near-negligible, practically “bias-free” level.
- Shared conclusion: Both sides agree bias must be actively measured, audited, and mitigated the debate is whether “near-zero, continuously monitored” bias should be called “completely free from bias” or not.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Ethical Framework Knowledge	30%	Strong grasp of ethical frameworks; cites real AI incidents accurately	Good grasp with minor inaccuracies	Basic grasp; limited supporting evidence	Weak or incorrect understanding of issues
Evidence-Based Argumentation	25%	Arguments grounded in real case studies with practical implications	Mostly grounded arguments	Few grounded examples used	Arguments unsupported by evidence
Argument Structure & Rebuttal	20%	Well-reasoned, logically structured arguments and rebuttals	Reasonably structured arguments	Weak structure or logic	Disorganized, illogical arguments
Delivery & Persuasion	15%	Persuasive, confident, respectful delivery	Clear, mostly confident delivery	Hesitant or unclear delivery	Unprepared or ineffective delivery
Team Engagement	10%	Active listening, effective rebuttal, strong teamwork	Reasonable engagement with team/opponents	Limited engagement in the debate	No meaningful engagement shown

Marks Evaluation:

Question No.	Ethical Framework Knowledge (30%)	Evidence-Based Argumentation (25%)	Argument Structure & Rebuttal (20%)	Delivery & Persuasion (15%)	Team Engagement (10%)	Total Marks (100)
Q1						
					Overall Total (100)	